香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 8: Dynamic Responses

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State-Space Models

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

where:

- $x(t) \in \mathbb{R}^n$ is the state at time t
- $u(t) \in \mathbb{R}^m$ is the input at time t
- $y(t) \in \mathbb{R}^p$ is the output at time t

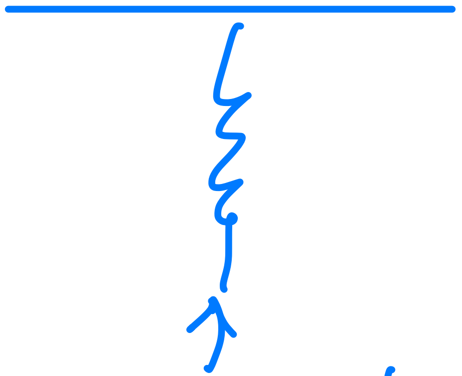
and

- $A \in \mathbb{R}^{n \times n}$ is the dynamics matrix
- $B \in \mathbb{R}^{n \times m}$ is the control matrix
- $C \in \mathbb{R}^{p \times n}$ is the sensor matrix

How do we determine the output y for a given input u?

Reminder: we will only consider single-input, single-output (SISO) systems, i.e., $u(t), y(t) \in \mathbb{R}$ for all times t of interest. $(m = p = 1)$

Impulse Response



only related to material itself

$$u \longrightarrow \boxed{\begin{array}{c}\dot{x} = Ax + Bu \\ y = Cx\end{array}} \longrightarrow y$$

zero initial condition: $x(0) = 0$

↳ will not affect the property of system!

Consider the input

$$u(t) = \delta(t - \tau) \qquad \text{unit impulse applied at } t = \tau$$

The system is *linear* and *time-invariant* (LTI), with zero I.C.:

impulse response

$$u(t) = \delta(t - \tau) \xrightarrow{x(0)=0;\ \text{LTI system}} y(t) = h(t - \tau)$$

The function h is the impulse response of the system.

Impulse Response

$$u \longrightarrow \boxed{\begin{array}{c} \dot{x} = Ax + Bu \\ y = Cx \end{array}} \longrightarrow y$$

zero initial condition: $x(0) = 0$

$$u(t) = \delta(t - \tau) \qquad \xrightarrow{x(0)=0;\ \text{LTI system}} \qquad y(t) = h(t - \tau)$$

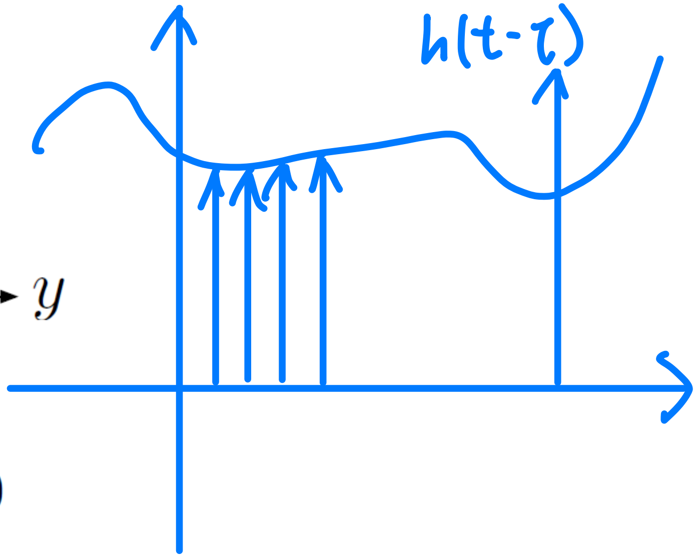
Questions to consider:

1. If we know h, how can we find the system's response to other (arbitrary) inputs?
2. If we don't know h, how can we determine it?

We will start with Question 1.

Impulse Response

$$u \longrightarrow \boxed{\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}} \longrightarrow y$$



zero initial condition: $x(0) = 0$

Question: If we know h, how can we find the system's response to other (arbitrary) inputs?

Recall the *sifting property* of the δ-function: for any function f which is "well-behaved" at $t = \tau$,

$$\int_{-\infty}^{\infty} f(t)\delta(t-\tau)\mathrm{d}t = f(\tau)$$

— any *reasonably regular* function can be represented as an integral of impulses!!

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

$$sX = AX + BU$$
$$Y = CX$$
$$(sI - A)X = BU$$
$$X = (sI - A)^{-1}BU$$
$$Y = CX$$
$$\Rightarrow \quad Y = C(sI - A)^{-1}BU$$
$$\frac{Y}{U} = C(sI - A)^{-1}B$$

Impulse Response

$$u \longrightarrow \boxed{\begin{array}{c} \dot{x} = Ax + Bu \\ y = Cx \end{array}} \longrightarrow y$$

zero initial condition: $x(0) = 0$

Question: If we know h, how can we find the system's response to other (arbitrary) inputs?

By the sifting property, for a general input $u(t)$ we can write

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\delta(t-\tau)\mathrm{d}\tau.$$

Now we recall the *superposition principle:* the response of a linear system to a sum (or integral) of inputs is the sum (or integral) of the individual responses to these inputs.

Why convolution can describe the output?

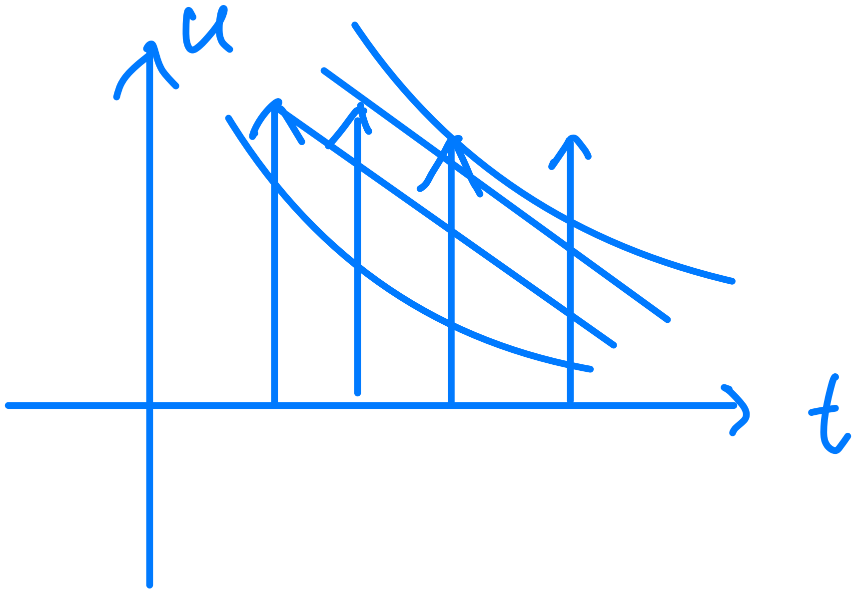


image processing

image $*$ kernel

$y =$

$$\sum f(\tau)\, h(t-\tau)$$

$$= \int_{-\infty}^{\infty} f(\tau)\, h(t-\tau)\, d\tau$$

$$= f * h$$

Impulse Response

$$u \longrightarrow \boxed{\begin{array}{c} \dot{x} = Ax + Bu \\ y = Cx \end{array}} \longrightarrow y$$

zero initial condition: $x(0) = 0$

The *superposition principle:* the response of a linear system to a sum (or integral) of inputs is the sum (or integral) of the individual responses to these inputs.

$$u(t) = \int_{-\infty}^{\infty} u(\tau)\delta(t-\tau)\mathrm{d}\tau \quad \longrightarrow \quad y(t) = \int_{-\infty}^{\infty} u(\tau)\underbrace{h(t-\tau)}_{\substack{\text{response to} \\ \delta(t-\tau)}}\mathrm{d}\tau$$

— the integral that defines $y(t)$ is a convolution of u and h.

Impulse Response

$$u \longrightarrow \boxed{\begin{array}{c}\dot{x} = Ax + Bu \\ y = Cx\end{array}} \longrightarrow y$$

zero initial condition: $x(0) = 0$

Conclusion so far: for zero initial conditions, the output is the convolution of the input with the system impulse response:

$$y(t) = u(t) \star h(t) = h(t) \star u(t) = \int_{-\infty}^{\infty} u(\tau)h(t-\tau)\mathrm{d}\tau$$

Q: Does this formula provide a *practical* way of computing the output y for a given input u?

A: Not directly (computing convolutions is not exactly pleasant), but ...we can use Laplace transforms.

Laplace Transforms and the Transfer Function

$$Y(s) = H(s)U(s), \qquad \text{where } H(s) = \int_{-\infty}^{\infty} h(\tau)e^{-s\tau}\mathrm{d}\tau$$

Given $u(t)$, we can find $U(s)$ using tables of Laplace transforms or `MATLAB`. But how do we know $h(t)$ [or $H(s)$]?

- Suppose we have a state-space model:

$$u \longrightarrow \boxed{\begin{array}{c}\dot{x} = Ax + Bu \\ y = Cx\end{array}} \longrightarrow y$$

Laplace Transforms and the Transfer Function

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- Suppose we have a state-space model:

$$u \longrightarrow \boxed{\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}} \longrightarrow y$$

In this case, we have an exact formula:

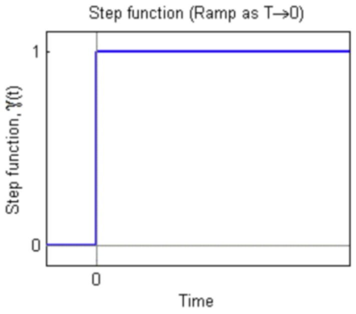
$$H(s) = C(Is - A)^{-1}B \qquad \text{(matrix inversion)}$$

$$h(t) = Ce^{At}B,\ t \geq 0^- \qquad \text{(matrix exponential)}$$

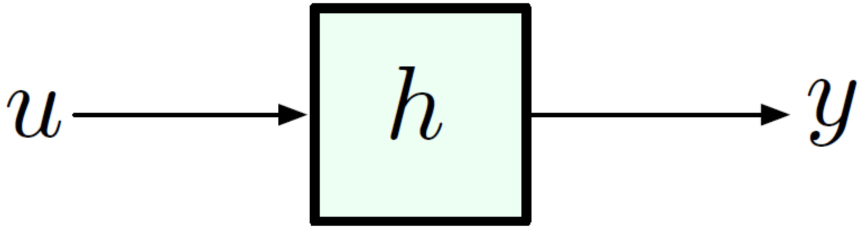
— will not encounter this until much later in the semester.

DC Gain

stable !



$u(t) = 1(t)$



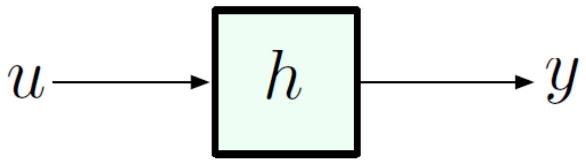
Definition: the steady-state value of the step response is called the *DC gain* of the system.

$$\text{DC gain} = y(\infty) = \lim_{t\to\infty} y(t) \qquad \text{for } u(t) = 1(t)$$

$u = \frac{1}{s}$

$\frac{Y}{u} = s$

Steady-State Value



$$u(t) = 1(t) \qquad U(s) = \frac{1}{s} \qquad \Longrightarrow \qquad Y(s) = \frac{H(s)}{s}$$

— can we compute $y(\infty)$ from $Y(s)$?

Let's look at some examples:

- $Y(s) = \dfrac{1}{s+a}$, $a > 0$ (pole at $s = -a < 0$)
  $y(t) = e^{-at} \quad \Longrightarrow \quad y(\infty) = 0$

  $\dfrac{s}{s+a}$

- $Y(s) = \dfrac{1}{s+a}$, $a < 0$ (pole at $s = -a > 0$)
  $y(t) = e^{-at} \quad \Longrightarrow \quad y(\infty) = \infty$

  cannot use FVT

- $Y(s) = \dfrac{1}{s^2+\omega^2}$, $\omega \in \mathbb{R}$ (poles at $s = \pm j\omega$, purely imaginary)
  $y(t) = \sin(\omega t) \quad \Longrightarrow \quad y(\infty)$ does not exist

- $Y(s) = \dfrac{c}{s}$ (pole at the origin, $s = 0$)
  $y(t) = c1(t) \quad \Longrightarrow \quad y(\infty) = c$

  only care about the right hand time/plane

In general, we have the *Final Value Theorem* (FVT), which states the following,

The Final Value Theorem (in Control): If all poles of $sY(s)$ are *strictly stable* or lie in the *open left half–plane* (OLHP), i.e., have $\mathrm{Re}(s) < 0$, then

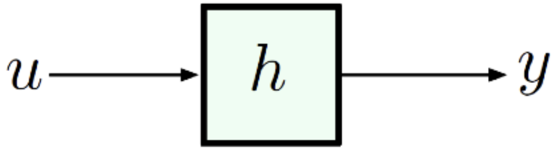
$$y(\infty) = \lim_{s \to 0} sY(s).$$

The Final Value Theorem (in Math): If $\lim_{t \to \infty} f(t)$ exists, i.e, it has a finite limit, then

$$\lim_{t \to \infty} f(t) = \lim_{s \to 0} sF(s),$$

where $F(s)$ is the one–sided Laplace transform of $f(t)$.

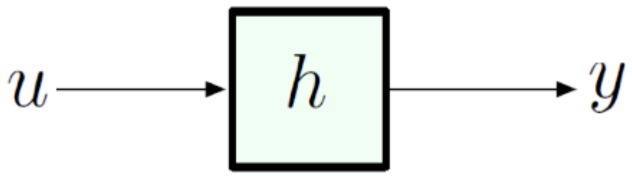
The Final Value Theorem



If we rerun the previous examples, hit $Y(s)$ with an s then pass $sY(s)$ to the limit,

- $Y(s) = \dfrac{1}{s+a}$, $sY(s) = \dfrac{s}{s+a}$. If $a > 0$, then $y(\infty) = 0$; if $a < 0$, FVT does not hold since unstable poles make $y(t)$ blow up as $t \to \infty$. The limit $y(\infty)$ does not exist in the first place.
- $Y(s) = \dfrac{1}{s^2+\omega^2}$, $sY(s) = \dfrac{s}{s^2+\omega^2}$, poles are purely imaginary (not in OLHP), FVT does not hold for the same reason that $y(\infty)$ does not exist in the first place.
- $Y(s) = \dfrac{c}{s}$, $sY(s) = c$, no poles or poles at infinity, so $y(\infty) = c$ is correctly given by FVT.

Back to DC Gain



$u(t) = 1(t) \qquad U(s) = \frac{1}{s} \qquad\qquad Y(s) = \frac{H(s)}{s}$

By Final Value Theorem, we may compute DC gain with ease when FVT holds. For example, system **unit step response** is nothing but

$$\boldsymbol{Y(s) = \frac{H(s)}{s}}.$$

☆ for step input use this

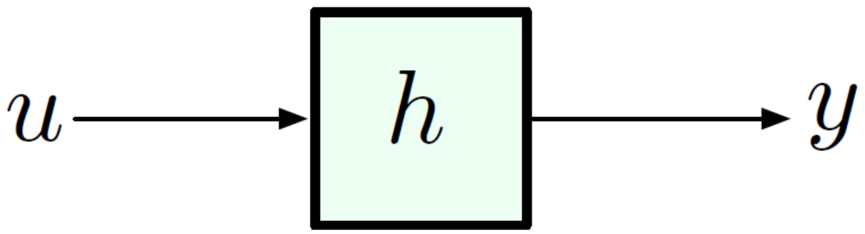
otherwise, use $\lim_{s\to 0} sY(s)$

If all poles of $sY(s) = H(s)$ are strictly stable, then by FVT

$$y(\infty) = \lim_{s\to 0} H(s).$$

> ***Important:*** *In general, we cannot pass the system transfer function $H(s)$ to the limit and claim the resulting limit is the steady–state value. The special feature of the above formula was that $Y(s)$ was the* ***unit step response****, where the input transform $\mathcal{L}\{1(t)\} = \frac{1}{s}$ would cancel s in the expression $sY(s)$ of FVT. If $Y(s)$ is not a unit step response, s will not be cancelled out by the Laplace transform of input $U(s) \neq \frac{1}{s}$.*

DC Gain



$$\begin{array}{lll} s^3 & 1 & 4 \\ s^2 & 2 & 5 \\ s & 1.5 & 0 \\ s^0 & 5 & 0 \end{array}$$

Example: Given system transfer function $\lim_{s \to 0} H(s) = \frac{3}{5}$

$$H(s) = \frac{s^2 + 5s + 3}{s^3 + 4s^2 + 2s + 5},$$

compute the DC gain of the system.

Solution: All the poles of $H(s)$ are strictly stable poles (we shall see this later using the *Routh—Hurwitz criterion*), so

$$y(\infty) = H(s)\Big|_{s=0} = \frac{3}{5}.$$

Review & Exercise

- How to determine y if we know x and h?
- How to determine h if we know x and y?
- What's zero initial condition?
- What's the sifting property of the δ-function?
- What's impulse response?
- Laplace transform and the transfer function?
- What's DC gain?
- What's the Final Value Theorem (FVT)?

- In-class exercise:

Question: Consider a linear time-invariant (LTI) system described by the following transfer function:

$$H(s) = \frac{10}{s^2 + 3s + 10}$$

1. **Calculate the DC gain of the system.**
2. **Interpret the result in the context of the system's steady-state response to a constant input.**

Ans:

DC Gain Calculation:

- The DC gain $H(0)$ is found by evaluating the transfer function at $s = 0$.
- Substitute $s = 0$ into the transfer function $H(s)$.

$$H(0) = \frac{10}{0^2 + 3(0) + 10} = \frac{10}{10} = 1$$

The DC gain of 1 means that if a constant input (e.g., a step input of magnitude 1) is applied to the system, the output will also reach a steady-state value of 1. This indicates that the system passes the constant input through unchanged in steady state.

Appendix

Matrix operation

Consider an LTI system described by state space equation

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{b}u(t)$$
$$y(t) = \boldsymbol{c}\boldsymbol{x}(t) + du(t).$$

no time/freq. domain

Take the Laplace transform with zero initial conditions:

$$s\boldsymbol{X}(s) = \boldsymbol{A}\boldsymbol{X}(s) + \boldsymbol{b}U(s) \qquad (2.20)$$
$$Y(s) = \boldsymbol{c}\boldsymbol{X}(s) + dU(s). \qquad (2.21)$$

Now a set of differential equations in the time domain becomes a set of algebraic equations in the frequency domain. There are a total of $n+1$ equations in (2.20)–(2.21) and we can use them to eliminate the n variables in $\boldsymbol{X}(s)$ to obtain an equation relating the input $U(s)$ and the output $Y(s)$. Linear algebra now gives a formal method of capturing this process. From (2.20), we get

$$\boldsymbol{X}(s) = (s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{b}U(s).$$

Plugging it into (2.21), we obtain

$$\boxed{\frac{Y(s)}{U(s)} = \boldsymbol{c}(s\boldsymbol{I} - \boldsymbol{A})^{-1}\boldsymbol{b} + d,}$$

i.e., the ratio of the Laplace transform of the output over that of the input is a fixed function independent of the input.

In this case, we have an exact formula:

$$H(s) = C(Is - A)^{-1}B \qquad \text{(matrix inversion)}$$

Note:

$$e^{-\alpha t}\sigma(t) \longleftrightarrow \frac{1}{s+\alpha}, \quad \mathrm{Re}\, s > -\alpha$$

$$h(t) = Ce^{At}B, \; t \geq 0^-$$

Proof of the Final Value Theorem (FVT)

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

The final value theorem (FVT) is one of several similar theorems used to relate frequency domain expressions to the time domain behaviour as time approaches infinity.

Proof

From the definition of the unilateral Laplace transform, we have,

$$L[x(t)] = X(s) = \int_0^\infty x(t)e^{-st}\,dt$$

can

Taking differentiation with respect to time on both sides, we get,

$$L[\frac{dx(\mathrm{t})}{dt}] = \int_0^\infty \frac{dx(\mathrm{t})}{dt}e^{-st}\,dt$$

Now, by the time differentiation property $[i.e.., \frac{dx(t)}{dt} \overset{LT}{\longleftrightarrow} sX(s) - x(0^-)]$of Laplace transform, we have,

$$L[\frac{dx(t)}{dt}] = \int_0^\infty \frac{dx(t)}{dt}e^{-st}\,dt = sX(s) - x(0^-)$$

By taking the $\lim_{s\to 0}$ on both sides of the above expression, we get,

$$\lim_{s\to 0}[\int_0^\infty \frac{dx(t)}{dt}e^{-st}\,dt] = \lim_{s\to\infty}[sX(s) - x(0^-)]$$

$$\Rightarrow \int_0^\infty \frac{dx(t)}{dt}\,dt = \lim_{s\to 0}[sX(s) - x(0^-)]$$

$$\Rightarrow [x(t)]_0^\infty = \lim_{s\to 0}[sX(\mathrm{s}) - x(0^-)]$$

$$\Rightarrow x(\infty) - x(0^-) = \lim_{s\to 0}[sX(s) - x(0^-)]$$

$$\Rightarrow x(\infty) = \lim_{s\to 0} sX(s)$$

Therefore, we have,

$$\lim_{t\to\infty} x(t) = x(\infty) = \lim_{s\to 0} sX(s)$$